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Screw Inserters

Abstract

A screw inserter for driving a screw into a vertebral body includes an inner element for engaging a screw portion of the screw and an outer element for engaging a coupling element of the screw. The outer element may include a threaded distal end. In addition, the screw inserter may include a button permitting the translation of the inner element with respect to the outer element. Further, the inserter may include an outer knob allowing the rotation of the inner element with respect to the outer element when moved to a first position. When the outer knob is moved to a second position it prevents the rotation of the inner element with respect to the outer element.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of the filing date of U.S. Provisional Patent Application No. 63/560,251 filed on Mar. 1, 2024, the disclosure of which is hereby incorporated by reference.

BACKGROUND OF THE INVENTION

[0002] Spinal screws, such as pedicle screws and lateral mass screws, have been used for many years by surgeons to correct degenerative spine diseases, spinal column injuries, and spinal column deformities. A typical screw procedure involves the implantation of screws into the different portions of different level vertebral bodies and their connection via a spinal rod or the like. This may or may not be done in concert with the implantation of an intervertebral implant, with the goal being the fusion of the vertebral bodies to each other.

[0003] Implantation of the screws into the vertebral bodies typically involves the use of a screw inserter to drive the screws into the vertebral bodies. These inserters may include inner and outer elements, with the inner element being designed to engage and drive the screw portion into the vertebral body, while the outer element attaches to the tulip or coupling element of the screw. Although these types of inserters generally operate for their intended purpose, their outer elements can catch on soft tissue during screw insertion, causing it to rotate and unthread from the screw. As a result, the screw is prone to accidentally uncoupling from the inserter during screw insertion.

[0004] Screw inserters have been designed to permit the locking of inner and outer elements to address the foregoing issues. While these involve different mechanisms to achieve the locking, they typically result in overly complicated and bulky designs. Other drawbacks may also include inner element tips that are susceptible to breaking due to uneven torque loading and actuation mechanisms that are susceptible to being actuated by mistake. These drawbacks can have particularly negative effects in cervical spine procedures, which tend to be more delicate than thoracic and lumbar procedures.

[0005] Thus, there is a need for a user-friendly screw inserter with a streamlined locking procedure that eliminates the known drawbacks.

BRIEF SUMMARY OF THE INVENTION

[0006] The present invention addresses the aforementioned drawbacks by providing a simple to use screw inserter that can lock inner and outer elements of the inserter both rotationally and translationally. The inserter includes one actuation mechanism for preventing/permitting rotational movement of the inner and outer elements with respect to each other and one for prevent/permitting translational movement. These are configured to provide a clear and simple operation for a user.

[0007] In accordance with an aspect of the present disclosure, a tool for driving a screw (e.g., a pedicle or lateral mass screw) into a vertebral body includes an inner element, including a distal end for engaging a screw portion of the screw, an outer element, including a threaded distal end for engaging a coupling element of the screw, a button, configured to permit translation of the inner element with respect to the outer element, and an outer knob. Movement of the outer knob to a first position permits rotation of the inner element with respect to the outer element and movement of the outer knob to a second position prevents rotation of the inner element with respect to the outer element. The tool also includes a locking button housed in an inner knob on the inner element.

[0008] In another aspect, the tool includes a ramped surface on the inner element configured to enable passive translation of the inner element with respect to the outer element.

[0009] In a different aspect, the inner element includes a more proximal reduced diameter and a more distal reduced diameter.

[0010] In a further aspect, the more proximal reduced diameter is configured to translate the inner element distally and the more distal reduced diameter is configured to facilitate passive disengagement of the threaded distal end from the screw.

[0011] In another aspect, the button includes a spring-loaded button.

[0012] In a different aspect, the tool includes an inner knob on the inner element including the button and a locking button.

[0013] In a further aspect, the tool includes an angled slot on the outer knob configured to attach to the inner knob and allow axial translation from the first position to the second position.

[0014] In another aspect, the locking button is configured to translate inward via an interface between the ramped surface and the outer knob, when the outer knob is moved distally into the second position.

[0015] In a different aspect, the tool includes a cut spline on the inner element that engages with the locking button in the second position.

[0016] In a further aspect, the outer knob translates proximally to disengage the locking button and rotationally decouple the inner element from the outer element.

[0017] In another aspect, the locking button includes a pawl configured to engage the cut spline on the inner element to prevent rotation in one or both directions.

[0018] In a different aspect, the pawl includes a spring-loaded pawl rotatably constrained to the locking button and configured to engage the cut spline when the locking button is depressed.

[0019] In a further aspect, the locking button includes a stop face above and below the spring-loaded pawl configured to prevent the spring-loaded pawl from moving further than intended.

[0020] In another aspect, the locking button includes a first pawl configured to translate and engage the cut spline, preventing rotation in one direction.

[0021] In yet another aspect, the locking button includes a second pawl that mirrors the first pawl about a vertical axis to the cut spline, preventing rotation in both directions.

[0022] In yet another aspect, the locking button includes a cantilever beam flexure configured to engage the cut spline and prevent rotation in one direction.

[0023] In yet another aspect, the locking button includes a cartwheel flexure, with a pivot point at an intersection of two connected flexures, configured to engage the cut spline and prevent rotation in one direction.

[0024] In yet another aspect, the locking button includes a small-length flexural pivot configured to engage the cut spline and prevent rotation in one direction.

[0025] In yet another aspect, the locking button includes a cross-axis flexural pivot, with a pivot point at an intersection of two disconnected flexures, configured to engage the cut spline and prevent rotation in one direction.

[0026] In yet another aspect, the locking button includes a beam flexure with teeth configured to engage the cut spline and prevent rotation in one or both directions.

[0027] In a further aspect, the locking button includes the beam flexure with at least one slot cut into the side of the beam flexure to create multiple flexures with a reduced width and stiffness, configured to engage the cut spline and prevent rotation in one direction.

[0028] In a different aspect, the locking button includes a set of left and right-side beam flexures, having at least one slot cut into the side of the beam flexure, that are intended to flex independently from each other and are configured to engage the cut spline and prevent rotation in one or both directions.

[0029] In accordance with yet another aspect of the present disclosure, a method of inserting a screw into a vertebral body includes threading an outer element of an inserter to a coupling element of a screw, engaging a distal end of an inner element of the inserter to a screw portion of the screw, translating an outer knob of the inserter distally to rotationally lock the inner element with respect to the outer element, driving the screw into a vertebral body via the inserter, translating the outer

knob of the inserter proximally to rotationally unlock the inner element from the outer element, and unthreading the outer element from the coupling element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] A more complete appreciation of the subject matter of the present invention and of the various advantages thereof can be realized by reference to the following detailed description in which reference is made to the accompanying drawings in which:

[0031] FIG. **1** is a perspective view of a locking screw inserter assembly in an unlocked and retracted position.

[0032] FIG. **2** is a perspective view of the locking screw inserter assembly of FIG. **1** in a locked and retracted position.

[0033] FIG. **3** is a perspective view of the locking screw inserter assembly of FIG. **1** in an unlocked and extended position.

[0034] FIG. **4** is an elevation view of an outer knob of the locking screw inserter assembly of FIG. **1** in an unlocked position.

[0035] FIG. **5** is an elevation view of the outer knob of the locking screw inserter assembly of FIG. **1** in a locked position.

[0036] FIG. **6** is a perspective view of an inner element of the locking screw inserter assembly of FIG. **1**.

[0037] FIG. **7** is a perspective view of the locking screw inserter assembly according to a different embodiment in a locked position, with certain portions thereof hidden.

[0038] FIG. **8** is an enlarged perspective view of the locking screw inserter assembly of FIG. **7** focusing on an inner knob.

[0039] FIG. **9** is a perspective view of the locking screw inserter assembly of FIG. **1** in an unlocked position, with certain portions thereof hidden.

[0040] FIG. **10** is an enlarged perspective view of the locking screw inserter assembly of FIG. **9** focusing on the inner knob.

[0041] FIG. **11** is a partial cross-sectional view focusing on a button and a locking button inside the inner knob according to an embodiment of the present invention.

[0042] FIG. **12** is a partial cross-sectional view focusing on the button and the locking button inside the inner knob, with a different inner element embodiment according to the present invention.

[0043] FIG. **13** is an elevation view of the button according to an embodiment of the present invention.

[0044] FIG. **14** is an enlarged elevation view of the locking button according to an embodiment of the present invention.

[0045] FIG. **15** is an elevation view of the locking button according to an embodiment of the present invention.

[0046] FIG. **16** is an elevation view of the locking button according to an embodiment of the present invention.

[0047] FIG. **17** is an elevation view of the locking button according to an embodiment of the present invention.

[0048] FIG. **18** is an enlarged elevation view of the locking button according to an embodiment of the present invention.

[0049] FIG. **19** is an elevation view of the locking button according to an embodiment of the present invention.

[0050] FIG. **20** is an elevation view of the locking button according to an embodiment of the present invention.

[0051] FIG. **21** is an elevation view of the locking button according to an embodiment of the present invention.

[0052] FIG. **22** is a perspective view of the locking button according to an embodiment of the present invention.

[0053] FIG. **23** is a perspective view of the locking button according to an embodiment of the present invention.

[0054] FIG. **24** is a perspective view of the locking button according to an embodiment of the present invention.

[0055] FIG. **25** is an enlarged perspective view of a portion of the locking button according to an embodiment of the present invention.

[0056] FIG. **26** is an enlarged perspective view of a portion of the locking button according to an embodiment of the present invention.

[0057] FIG. **27** is an enlarged perspective view of a portion of the locking button according to an embodiment of the present invention.

[0058] FIG. **28** is a perspective view of the locking button according to an embodiment of the present invention.

[0059] FIG. **29** is a perspective view of the locking button according to an embodiment of the present invention.

[0060] FIG. **30** is an elevation view of the locking button according to an embodiment of the present invention.

[0061] FIG. **31** is a perspective view of the locking button of FIG. **30**.

[0062] FIG. **32** is a partial cross-sectional view focusing on the locking button of FIG. **30** inside the inner knob.

DETAILED DESCRIPTION

[0063] In describing the preferred embodiments of the invention, specific terminology will be used for the sake of clarity. However, the invention is not intended to be limited to any specific terms used herein, and it is to be understood that each specific term includes all technical equivalents, which operate in a similar manner to accomplish a similar purpose. In the drawings and in the description which follows, the term “proximal” refers to the end of the instrument, or portion thereof, which is closest to the operator in use, while the term “distal” refers to the end of the instrument, or portion thereof, which is farthest from the operator in use.

[0064] Referring to FIGS. **1-3**, a screw inserter (“inserter”) **200** according to an embodiment of the present invention includes an inner element **202**, an outer element **206**, a button **210** and an outer knob **212**. Outer element **206** is incased in an outer case **207**. As better shown in FIGS. **8** and **10-12**, inserter **200** may also include an inner knob **226** with a locking button **300**. With particular reference to FIGS. **2** and **3**, inner element **202** includes a distal end **204** for engaging a screw portion **101** of a screw **100** (e.g., pedicle screw or lateral mass screw) and outer element **206** includes a threaded distal end **208** for engaging a coupling element **103** of screw **100**. Button **210** is housed within inner knob **226** and a portion of inner element **202** is designed to extend through the button. As will be described in more detail below, depressing button **210** permits translation of inner element **202** with respect to outer element **206** of inserter **200** from a retracted position **400** into an extended position **500**.

[0065] As shown in FIG. **6**, in addition to distal end **204**, inner element **202** includes three reduced diameters—a first more proximal reduced diameter **218**, a second more proximal reduced diameter **219**, and a more distal reduced diameter **220**. First more proximal reduced diameter **218** and second more proximal reduced diameter **219** are separated by ramped surface **222** and are configured to lock inner element **202** in two different translational positions. As shown in FIG. **13**, button **210** includes a first inner width **232** that is less than the diameter of inner element **202**, which allows inner element to remain in retracted position **400** with button contacting second more proximal reduced diameter **219**. Depressing button **210** allows inner element **202** to shift to a second inner

width **234** of button, that is greater than the diameter of inner element, which allows inner element to translate into extended position **500** with first more proximal reduced diameter **218** contacting button. As shown in FIGS. **7** and **9**, inner element **202** can only translate into these positions when button **210** is depressed, and reduced diameters **218**, **219** move into second inner width **234** of button. As shown in FIGS. **11** and **12**, in one embodiment of the present invention, button **210** is biased to the position in which inner element **202** engages width **232** via a spring **203**.

[0066] As shown in FIGS. **4-5** and **7-10**, movement of outer knob **212** to first position **214** (FIG. **4**) permits rotation of inner element **202** with respect to outer element **206**, i.e., an unlocked position of inserter **200**. Moving outer knob **212** into second position **216** (FIG. **5**) prevents rotation of inner element **202** with respect to outer element **206**, i.e., a locked inserter **200**. Outer knob **212** includes angled slot **230** configured to permit extension of inner knob **226** to slide therein and allow axial translation from first position **214** to second position **216**.

[0067] In addition to button **210**, inner knob **226** also includes locking button **300**, which is configured to be depressed or translated inward via an interface between a ramped inner surface of the outer knob and outer knob **212** when outer knob is moved distally into second position **216**. When locking button **300** is depressed, a portion of the locking button engages with a cut spline **228** on inner element **202** in second position **216**. This rotationally locks the inner and outer elements to each other. To unlock inserter **200**, outer knob **212** translates proximally back to first position **214** to disengage locking button **300** and decouple inner element **202** from outer element **206** rotationally.

[0068] Locking buttons according to the present invention may exhibit various configurations, most notable in connection with the manner in which the button engages with a cut spline. Reference will now be made to various embodiment splines and locking buttons. Wherever possible, the same or like reference numbers will be used throughout the drawings to refer to the same or like features within a different 100-series of numbers.

[0069] Beginning with FIG. **14**, locking button **600** and cut spline **628** are shown. Locking button **600** includes a spring-loaded pawl **636** with a tooth **637** that engages cut spline **628** to prevent rotation of inner element **602** in both rotational directions. Locking button **600** includes stop faces **642** that constrain spring-loaded pawl **636**, which is rotatably attached to locking button **600** at pivot point **641**. A spring **635** biases pawl **636** such that engagement of tooth **637** with spline **628** is ensured. Locking button springs (not shown) allow locking button **600** to depress to engage cut spline **628** when outer knob (not shown) is translated distally into second position (not shown).

[0070] As shown in FIG. **15**, another embodiment of the present invention includes locking button **700** comprising a first pawl **738** configured to translate and engage cut spline **728** to prevent rotation. In particular, first pawl **738** includes an edge **739** that engages cut spline **728** to prevent rotation of inner element **702** in at least one direction. Spring **735** biases first pawl **738** such that engagement of edge **739** with cut spline **728** is ensured when locking button **700** is depressed and cut spline moves upward into contact with first pawl. The moveability of first pawl **738** permits engagement even when cut spline **728** is not in a specific orientation with respect to first pawl, as well as permitting rotation of inner element **702** in one direction. Locking button **700** depresses in a similar fashion to locking button **600** discussed above.

[0071] FIG. **16** depicts another embodiment locking button **800** that includes first and second pawls **838**, **840** that are spaced apart from each other, but which can both engage spline **828**. First pawl **838** and second pawl **840** are configured to translate vertically and engage cut spline **828**, much like in connection with pawl **738**. However, the inclusion of two pawls in this embodiment prevents rotation in both directions. First pawl **838** and second pawl **840** each include edges **839** that engage cut spline **828** at the same time when locking button **800** is depressed via locking button springs **855** to prevent rotation in both directions. In particular, locking button **800** includes first pawl **838** and second pawl **840** attached to locking button **800** via springs **835** that bias first pawl **838** and second pawl **840** such that engagement of edge **839** with cut spline **828** is ensured via vertical

translation when first and second pawls are moved towards cut spline. Locking button **800** depresses in a similar fashion to locking button **600** discussed above.

[0072] As shown in FIG. **17**, another embodiment locking button **900** includes cantilever beam flexure **944**. Cantilever beam flexure **944** engages cut spline **928** on inner element **902** to prevent rotation in one direction when locking button **900** is depressed. In particular, locking button **900** includes cantilever beam flexure **944** having tooth **937** configured to engage cut spline **928** to prevent rotation in one direction. Cantilever beam flexure **944** is rotatably attached to locking button **900** at pivot point **941** configured to contact cut spline **928** with tooth **937** when locking button is depressed and cut spline moves upward. The design of cantilever beam flexure **944** is such that tooth **937** is biased towards spline **928** without the need for a separate spring element. Locking button **900** depresses in a similar fashion to locking button **600** discussed above.

[0073] Another embodiment locking button **1000** is shown in FIG. **18** and includes a similar embodiment cantilever beam flexure **1044**. Cantilever beam flexure **1044** engages cut spline **1028** on inner element **1002** to prevent rotation in both directions when locking button **1000** is depressed. In particular, cantilever beam flexure **1044** includes teeth **1037** configured to engage cut spline **1028** to prevent rotation in both directions. Cantilever beam flexure **1044** otherwise operates in a similar fashion to flexure **944** discussed above. Locking button **1000** depresses in a similar fashion to locking button **600** discussed above.

[0074] In other embodiments, locking button may include a one-piece construction including an integral portion that engages the cut spline to prevent rotation in one or both directions. Such locking buttons depress in a similar fashion to locking button **600** discussed above via locking button springs. The remaining embodiments shown in the figures exhibit such a design.

[0075] As shown in FIG. **19**, another embodiment of locking button **1100** includes a one-piece construction with a small-length flexural pivot **1154**. Small-length flexural pivot **1154** engages cut spline (not shown) preventing rotation in one direction when locking button **1100** is depressed. In particular, small-length flexural pivot **1154** includes tooth **1137** configured to engage cut spline to prevent rotation in one direction. Small-length flexural pivot **1154** is rotatably attached to locking button **1100** at pivot or flex point **1141**. Small-length flexural pivot **1154** is made of a flexible material allowing for bending as locking button **1100** is depressed to engage cut spline.

[0076] As shown in FIG. **20**, another embodiment of locking button **1200** includes a one-piece construction with a cross-axis flexural pivot **1248**, that has pivot or flex point **1241** at an intersection of two disconnected flexures. Cross-axis flexural pivot **1248** engages cut spline (not shown) and prevents rotation in one direction when locking button **1200** is depressed. In particular, cross-axis flexural pivot **1248** includes edge **1239** configured to engage cut spline to prevent rotation in one direction. Cross-axis flexural pivot **1248** is made of a flexible material allowing for bending as locking button **1200** is depressed to engage cut spline.

[0077] As shown in FIG. **21**, another embodiment of locking button **1300** includes a one-piece construction with a cartwheel flexure **1346**, that has pivot or flex point **1341** at an intersection of two connected flexures. Cartwheel flexure **1346** engages cut spline **1328**, preventing rotation in one direction when locking button **1300** is depressed. In particular, cartwheel flexure **1346** includes edge **1339** configured to engage cut spline **1328** to prevent rotation in one direction. Cartwheel flexure **1346** is made out of a flexible material allowing flexure to bend and rotate around pivot point **1341** to engage cut spline **1328** as locking button **1300** is depressed via locking button springs **1355**.

[0078] As shown in FIG. **22**, another embodiment of locking button **1400** includes beam flexure **1450** with tooth **1437** that engages cut spline (not shown) when locking button **1400** is depressed to prevent rotation in both directions. Beam flexure **1450** is made out of a flexible material allowing beam flexure to bend for easier engagement of tooth **1437** with cut spline as locking button **1400** is depressed.

[0079] As shown in FIG. **23**, another embodiment of locking button **1500** includes beam flexure

1550 with tooth **1537** having at least one slot **1552** cut into the side of beam flexure. Slots **1552** create multiple flexures with a reduced width and stiffness intended to increase the flexibility of beam flexure **1550** when engaging the cut spline (not shown). When locking button **1500** is depressed, tooth **1537** will engage the cut spline. Exact alignment between tooth **1537** and cut spline is not required to prevent rotation because beam flexure **1550** may be made of flexible material allowing for easier engagement of tooth by cut spline.

[0080] As shown in FIG. **24**, another embodiment of locking button **1600** includes beam flexure **1650** with teeth **1637** having at least one slot **1652** cut into the side of beam flexure. Slots **1652** create multiple flexures with a reduced width and stiffness intended to increase the flexibility of beam flexure **1650** when engaging cut spline (not shown). This design is similar to that of FIG. **23**, albeit with more teeth for engaging the cut spline.

[0081] Additional embodiment of locking buttons **1700**, **1800**, **1900** are shown in FIGS. **25-27**. These each include a set of left and right-side beam flexures **1756a**, **1756b**; **1856a**, **1856b**; **1956a**, **1956b** having at least one slot **1752**, **1852**, **1952** cut into the side of each beam. Each beam flexure **1756a**, **1756b**; **1856a**, **1856b**; **1956a**, **1956b** is intended to flex independently from each other to engage cut spline (not shown), preventing rotation in one or both directions. In particular, beam flexures **1756a**, **1756b**; **1856a**, **1856b**; **1956a**, **1956b** include teeth **1737**, **1837**, **1937** that are configured to engage cut spline when locking button **1700**, **1800**, **1900** is depressed. The shape of teeth **1737**, **1837**, **1937** in locking button **1700**, **1800**, **1900** may vary to prevent rotation in both directions. Exact alignment between teeth **1737**, **1837**, **1937** and cut spline is not required to prevent rotation because beam flexures **1756a**, **1756b**; **1856a**, **1856b**; **1956a**, **1956b** are made of flexible material allowing for easier engagement of teeth by cut spline.

[0082] As shown in FIGS. **28** and **29**, further embodiment locking button **2000** **2100** include a one-piece construction with beam flexures **2050**, **2150** including teeth **2037**, **2137** configured to flex in order for teeth to meet peak-to-peak with cut spline (not shown) to provide a seamless lock. The in-plane width of beam flexure **2050**, **2150** may be tailored to achieve the desired level of flexibility. In some embodiments, locking button **2000**, **2100** is made of a flexible material that allows beam flexures **2050**, **2150** to bend as locking button is depressed and teeth **2037**, **2137** engage cut spline.

[0083] As shown in FIGS. **30-32**, another embodiment locking button **2200** also includes a one-piece construction having an opening with teeth **2237** configured to meet peak-to-peak with cut spline **2228**. Cut spine **2228** includes curved protrusions configured to fit seamlessly between teeth **2237** of locking button **2200**. This provides a seamless lock upon depression of locking button. As shown in FIGS. **30** and **31**, locking button **2200** includes three curved teeth **2237** arranged along an upper end of the opening. Such a curvature of teeth **2237** and cut spine **2228** may prevent rotation in one or both directions.

[0084] In accordance with another aspect of the present disclosure, a method of inserting screw into vertebral body pedicle includes threading screw **100** onto inserter **200** by mating coupling element of screw **103** with threaded distal end of inserter **208**, locking inserter by translating outer knob **214** distally, driving screw into a vertebral body, unlocking inserter by translating outer knob proximally, and unthreading inserter from screw.

[0085] As shown in FIG. **3**, extended position **500** of inner element **202** facilitates threaded distal end **208** in threading coupling element **103** of screw **100** by engaging coupling element and then proximally pushing screw until threaded distal end contacts coupling element. After extended position **500** engages coupling element **103** of screw **100**, according to an embodiment of the present invention with a straight surface **224** on inner element **202**, button **210** is depressed to disengage coupling element **103** of screw **100** and enable inner element to retract into inserter **200**. Depressing button **210** advances inserter **200** distally and threads inserter into coupling element **103** of screw **100**. When inserter **200** is fully threaded into coupling element **103** of screw **100**, button **210** will move outward and engage a flat faced flange of the more distal reduced diameter **218** which facilitates disengagement of threaded distal end **208** of inner element **202** from screw

100 during removal of inserter **200**. After inserter **200** is fully engaged with screw **100**, outer knob **212** is translated distally to put inserter into a second position **216**, rotationally locking the inserter. After driving screw portion **101** of screw **100** into a vertebral body, outer knob **212** is translated proximally into a first position **214** to rotationally unlock the inserter **200**. Then inserter **200** is unthreaded from the fully seated screw **100** and inner element **202** is pulled out of engagement with coupling element **103** of screw.

[0086] According to another embodiment of the present invention, inner element **202** includes a ramped surface **222** that enables passive translation of inner element with respect to the outer element **206** when button **210** is depressed. Ramped surface **222** allows inner element **202** to retract into inserter **200** while pushing screw **100** onto inserter allowing a passive transition from extended position **500** to retracted position **400**. Extended position **500** allows screw **100** to be proximally pushed to threaded distal end **208** of inserter **200**, and inner element **202** returns to retracted position **400**.

[0087] It is to be understood that a proximal end of inner element **202** is to be attached to a separate driving element for use during screw insertion into the vertebral body. Indeed, the proximal end is shown having a typical structure for attachment to such devices. For instance, a torque handle and/or surgical drill/driver could be coupled with this structure and utilized during a surgical procedure involving the driver. It is contemplated that other driving devices could be utilized and that the proximal end of inner element **202** could employ any connection structure suitable for such use.

[0088] Although the invention herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. For example, features described in relation to one particular embodiment may be combined with features of other embodiments described herein. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined in the appended claims.

Claims

1. A tool for driving a screw into a vertebral body, comprising: an inner element including a distal end for engaging a screw portion of the screw; an outer element including a threaded distal end for engaging a coupling element of the screw; a button, wherein depressing the button permits translation of the inner element with respect to the outer element; and an outer knob, wherein movement of the outer knob to a first position permits rotation of the inner element with respect to the outer element and movement of the outer knob to a second position prevents rotation of the inner element with respect to the outer element.
2. The tool of claim 1, further comprising a ramped surface on the inner element configured to enable passive translation of the inner element with respect to the outer element.
3. The tool of claim 1, wherein the inner element includes a more proximal reduced diameter and a more distal reduced diameter, the more proximal reduced diameter is configured to translate the inner element distally and the more distal reduced diameter is configured to facilitate passive disengagement of the threaded distal end from the screw.
4. The tool of claim 1, wherein the button includes a spring-loaded button.
5. The tool of claim 2, further comprising an inner knob on the inner element including the button and a locking button.
6. The tool of claim 5, further comprising an angled slot on the outer knob configured to attach to the inner knob and allow axial translation from the first position to the second position.
7. The tool of claim 5, wherein the locking button is configured to translate inward via an interface between the ramped surface and the outer knob, when the outer knob is moved distally into the second position.

- 8.** The tool of claim 5, further comprising a cut spline on the inner element that engages with the locking button in the second position.
 - 9.** The tool of claim 5, wherein the locking button includes a pawl configured to engage the cut spline on the inner element to prevent rotation in one or both directions.
 - 10.** The tool of claim 5, wherein the locking button includes one or more teeth configured to engage the cut spline on the inner element to prevent rotation in one or both directions.
 - 11.** The tool of claim 5, wherein the locking button includes a cantilever beam flexure configured to engage the cut spline and prevent rotation in one direction.
 - 12.** The tool of claim 5, wherein the locking button includes a cartwheel flexure, with a pivot point at an intersection of two connected flexures, configured to engage the cut spline and prevent rotation in one direction.
 - 13.** The tool of claim 5, wherein the locking button includes a small-length flexural pivot configured to engage the cut spline and prevent rotation in one direction.
 - 14.** The tool of claim 5, wherein the locking button includes a cross-axis flexural pivot, with a pivot point at an intersection of two disconnected flexures, configured to engage the cut spline and prevent rotation in one direction.
 - 15.** The tool of claim 5, wherein the locking button includes a beam flexure with teeth configured to engage the cut spline and prevent rotation in one or both directions.
 - 16.** A method of inserting a screw into a vertebral body, comprising: threading an outer element of an inserter to a coupling element of a screw; engaging a distal end of an inner element of the inserter to a screw portion of the screw; translating an outer knob of the inserter distally to rotationally lock the inner element with respect to the outer element; driving the screw into a vertebral body via the inserter; translating the outer knob of the inserter proximally to rotationally unlock the inner element from the outer element; and unthreading the outer element from the coupling element.
 - 17.** The method of claim 16, wherein the inner element includes a more proximal reduced diameter and a more distal reduced diameter, and the more proximal reduced diameter translates the inner element distally and the more distal reduced diameter facilitates passive disengagement of the threaded distal end from the screw.
 - 18.** The method of claim 16, further comprising depressing a button within the inserter to translate the inner element with respect to the outer element using a ramped surface on the inner element.
 - 19.** The method of claim 18, further comprising moving the outer knob distally to lock the inserter using a locking button within the inserter, wherein the locking button translates inward via an interface between the ramped surface and the outer knob.
 - 20.** The method of claim 19, further comprising proximally translating the outer knob to disengage the locking button and rotationally decouple the inner element from the outer element.
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